

A Comparative Study of Rule-Based, Machine Learning,
and Hybrid Dynamic Pricing Models for Stable and
Customer-Centric Pricing in Retail SMEs

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Declaration

I hereby declare that this dissertation entitled *A Comparative Study of Rule-Based, Machine Learning, and Hybrid Dynamic Pricing Models for Stable and Customer-Centric Pricing in Retail SMEs* is entirely my own work and has been submitted in partial fulfilment of the requirements for the award of Master of Science (MSc) in Computer Science.

This dissertation was completed for the MSc Computing Project module (October 2025). Except where appropriately acknowledged and referenced, the work presented in this dissertation is original and has not been submitted previously, either in whole or in part, for any other degree, diploma, or academic qualification at this or any other institution.

All sources of information, datasets, and supporting materials used in this research have been appropriately acknowledged and referenced in accordance with academic standards.

Abstract

This study examines the comparative performance of rule-based, machine learning-based, and hybrid dynamic pricing approaches within retail small and medium-sized enterprise (SME) contexts. Existing research on dynamic pricing frequently prioritises revenue optimisation in data-rich environments while giving less attention to pricing consistency and the operational constraints faced by SMEs. In retail settings characterised by repeated customer interactions, excessive price variation may influence trust and perceived fairness, making pricing stability an important evaluation dimension alongside financial performance.

The study adopts a computational experimental design using a publicly available retail dataset transformed into a structured product-level time series. Three pricing approaches are implemented within a consistent simulation framework: a rule-based system using predefined heuristics, a machine learning model that selects prices according to predicted revenue, and a hybrid model combining optimisation with constraints on price changes. All approaches are evaluated using identical data and performance metrics, ensuring that observed differences arise from pricing logic rather than implementation inconsistencies.

The results show that the machine learning approach achieves the highest revenue performance but also exhibits the greatest pricing variability. The rule-based approach produces moderate outcomes across both dimensions without fully optimising either objective. The hybrid approach generates the most stable pricing behaviour, significantly reducing price variation while maintaining comparatively competitive revenue outcomes. Statistical testing confirms that differences in pricing stability are significant, whereas revenue differences remain relatively moderate.

The study contributes to the literature by providing a structured comparative evaluation of dynamic pricing approaches within an SME-focused context and by incorporating pricing stability as a core evaluation criterion. The findings demonstrate that pricing performance should not be assessed solely through optimisation outcomes and highlight the practical suitability of hybrid pricing approaches for retail SMEs operating under data and resource constraints.

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Chapter 1: Introduction

1.1 Introduction

Pricing is a central element of retail strategy, influencing revenue generation, competitive positioning, and customer purchasing behaviour. In digital retail environments, pricing decisions have increasingly shifted from static policies towards adaptive systems capable of responding to changing market conditions. Advances in data analytics and computational capability have enabled the development of dynamic pricing models that adjust prices according to demand signals and market information. Within academic research, dynamic pricing is commonly framed as a sequential decision process in which firms update pricing policies based on new information and observed outcomes (den Boer and Keskin, 2022).

Although dynamic pricing is often associated with improved financial performance, its application within retail settings introduces additional considerations. Retail markets involve repeated customer interactions and visible price fluctuations, meaning that pricing decisions may also influence customer perceptions of fairness and consistency. Research indicates that consumers can respond negatively when price changes appear inconsistent or unfair (Diao, Harutyunyan and Jiang, 2023). These concerns are particularly relevant for retail small and medium-sized enterprises (SMEs), where pricing decisions may carry stronger reputational implications. Against this background, this dissertation examines how alternative dynamic pricing approaches perform within retail SME environments.

1.2 Background to the Study

Dynamic pricing research has evolved through contributions from operations research, economics, and data-driven decision modelling. Early approaches relied on deterministic models in which demand was treated as a stable function of price, while later research introduced stochastic and adaptive learning methods to address uncertainty. Contemporary studies frequently frame pricing as an optimisation problem under uncertainty, with performance commonly evaluated through revenue or profit-based measures (Kopalle *et al.*, 2023).

Within this literature, three broad categories of pricing approaches are commonly identified. Rule-based models apply predefined pricing rules and prioritise interpretability and managerial oversight. Machine learning-based models use historical data to estimate demand relationships or generate pricing policies through iterative optimisation procedures (Fan, Guo and Yu, 2024). Hybrid approaches combine structured constraints with adaptive components to balance responsiveness and control.

However, these approaches are often evaluated in data-rich environments supported by advanced analytical infrastructure. Retail SMEs frequently operate under more constrained conditions, including limited historical data and restricted technical capacity (Gouveia and Mamede, 2022). Consequently, pricing systems developed for large-scale environments may not be directly suitable for smaller retail firms. This context forms the basis for the research problems addressed in this dissertation.

1.3 Problem Statements

Despite advances in dynamic pricing research, much of the literature evaluates pricing approaches primarily through optimisation-oriented criteria such as revenue maximisation or policy efficiency (Kopalle *et al.*, 2023). While these measures provide analytical clarity, they may not fully capture dimensions that are particularly relevant in retail environments, including the stability of pricing behaviour over time.

A further issue concerns the assumptions embedded within learning-based pricing models. Many machine learning approaches assume sufficient data availability, reliable feedback, and organisational capacity to support continuous model refinement. Retail SMEs, however, often operate under more constrained conditions, including limited analytical expertise and smaller datasets (Gouveia and Mamede, 2022). As a result, the performance of models developed in data-rich contexts may not transfer directly to SME environments.

Finally, rule-based, machine learning-based, and hybrid pricing approaches are frequently examined separately and under differing assumptions. Comparative evaluation of these approaches within a consistent retail SME setting, using criteria that extend beyond revenue outcomes, remains limited. Consequently, it is not always clear how these pricing paradigms differ in terms of stability and suitability for smaller retail organisations.

1.4 Aim

This dissertation aims to comparatively evaluate rule-based, machine learning-based, and hybrid dynamic pricing models within retail SME contexts. The study assesses these approaches in

terms of **revenue performance** and **pricing stability** in order to determine their relative suitability within constrained retail environments.

1.5 Objectives

To achieve this aim, the dissertation pursues the following objectives:

1. To examine how rule-based, machine learning-based, and hybrid dynamic pricing approaches are conceptualised within retail contexts.
2. To implement these pricing models using retail data under consistent experimental conditions.
3. To evaluate and compare their performance in terms of revenue outcomes and pricing stability.

1.6 Research Question

The study is guided by the following research question:

How do rule-based, machine learning-based, and hybrid dynamic pricing models compare in terms of pricing stability and revenue performance within retail SME contexts?

1.7 Significance of the Study

This study contributes to the evaluation of dynamic pricing models within retail environments by extending assessment beyond optimisation-based outcomes. Existing research frequently prioritises revenue or profit maximisation as primary indicators of performance (Kopalle *et al.*, 2023). While such measures remain important, they may not fully reflect the practical implications of pricing behaviour within smaller retail settings. By incorporating **pricing**

stability as an additional evaluative dimension, this research supports a more balanced assessment of dynamic pricing approaches.

The study also has practical relevance for retail SMEs operating under constrained data and resource conditions. Comparative evidence within a retail SME setting may assist managers in assessing the suitability of alternative pricing approaches without assuming that methods developed for larger firms are directly transferable.

1.8 Scope and Purpose of the Study

This dissertation focuses on the comparative evaluation of rule-based, machine learning-based, and hybrid dynamic pricing models within retail SME contexts. The study uses retail data under controlled experimental conditions and does not involve live commercial deployment or real-time market experimentation. Its purpose is evaluative rather than prescriptive, focusing on comparative model behaviour rather than the development of a novel pricing algorithm.

Within this study, **pricing stability is used as a proxy for customer-oriented pricing behaviour** rather than a direct measure of customer perception. Stability is defined through the magnitude and frequency of price changes over time. Previous research suggests that price volatility can influence perceptions of fairness and trust, particularly in repeated retail interactions (Diao, Harutyunyan and Jiang, 2023). However, this study does not directly measure behavioural responses and instead evaluates observable pricing patterns.

1.9 Dissertation Structure

This dissertation is organised into six chapters. Chapter 1 introduces the research context, problem statements, aim, objectives, and research question. Chapter 2 reviews the literature on

dynamic pricing approaches. Chapter 3 outlines the research methodology and model implementation process. Chapter 4 presents the comparative findings of the pricing models. Chapter 5 discusses the results in relation to the research question and existing literature. Finally, Chapter 6 concludes the study and identifies limitations and future research directions.

1.10 Summary

This chapter has introduced the research context and outlined the key issues associated with evaluating dynamic pricing models within retail SME environments. It has defined the research aim, objectives, and research question, while clarifying the study's evaluative focus on revenue performance and pricing stability. The chapter has also established the scope of the research and provided an overview of the dissertation structure, forming the foundation for the literature review presented in Chapter 2.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews the literature on dynamic pricing within retail contexts and small and medium-sized enterprises (SMEs). Its purpose is to examine how alternative pricing approaches have been conceptualised and evaluated across research traditions, while identifying limitations that inform the comparative framework adopted in this dissertation. A structured narrative synthesis is used because the literature spans heterogeneous modelling paradigms, evaluative criteria, and disciplinary assumptions that cannot be meaningfully consolidated statistically.

Dynamic pricing research draws from operations research, economics, and machine learning, each embedding different assumptions regarding demand behaviour, data availability, experimentation tolerance, and organisational capability. These assumptions are particularly significant in SME retail environments characterised by constrained data conditions and heightened reputational sensitivity. The chapter compares deterministic, learning-based, hybrid, behavioural, and organisational perspectives, focusing on how pricing models are framed, evaluated, and positioned for practical implementation.

2.2 Conceptual Foundations of Dynamic Pricing

The conceptual foundations of dynamic pricing frame pricing as an optimisation or control problem in which demand responds systematically to price adjustments. Deterministic, stochastic, and dynamic formulations share assumptions regarding demand stability, observability, and parameter reliability.

Analytical price–demand models assume that demand can be represented as a stable function of price, enabling revenue-maximising solutions through optimisation (Kayikci *et al.*, 2022). While mathematically tractable, these models simplify behavioural heterogeneity and contextual variation. Stochastic extensions incorporate probabilistic demand processes to address uncertainty (Diao, Harutyunyan and Jiang, 2023), though uncertainty is often treated as variation around stable demand structures rather than as structural behavioural change.

Elasticity-based approaches similarly assume that consumer responsiveness can be reliably estimated and used to guide pricing decisions (den Boer and Keskin, 2022). However, research on reference price effects and loss aversion suggests that demand responsiveness may be path-dependent and unstable over time. Sequential and control-based formulations attempt to address temporal interdependence by recognising that current prices influence future system states (Kigo, Omondi and Omolo, 2023). Yet these models often assume sufficient data and analytical infrastructure to support continuous policy updates.

Theoretical syntheses consolidate these approaches into broader optimisation frameworks (Yang and Zhang, 2022), while operational extensions integrating inventory or capacity constraints increase realism but also reliance on accurate parameter estimation (Cohen, Elmachoub and Lei, 2022). Overall, the conceptual literature prioritises analytical tractability over behavioural and organisational realism.

2.3 Deterministic and Rule-Based Pricing Approaches

Deterministic and rule-based pricing approaches prioritise transparency, predictability, and managerial control over adaptivity. Rather than treating pricing as a learning process, these models govern price changes through predefined rules or optimisation logic.

Rule-based pricing commonly relies on heuristics derived from managerial knowledge or domain expertise (Das *et al.*, 2024). These heuristics determine how prices respond to observable signals such as demand, inventory, or timing conditions. Threshold-based and staged pricing strategies further limit adjustments to discrete events, reducing volatility and simplifying operations (Immadisetty, 2025). However, the effectiveness of such systems depends heavily on stable environmental conditions and appropriate threshold calibration.

Deterministic optimisation models formalise non-adaptive pricing by deriving optimal prices under assumed demand and cost structures (Deng, 2023). Although analytically coherent, their outcomes depend on accurate parameter estimation, which may be difficult in uncertain retail environments. Practical baseline strategies described in the literature favour simple and deployable pricing rules that generate incremental gains without significant analytical complexity (Nouri-Harzvili and Hosseini-Motlagh, 2023).

Retail-focused studies also frame pricing rigidity as a safeguard against customer dissatisfaction and reputational risk (Chen *et al.*, 2022). Consequently, deterministic approaches often prioritise stability and interpretability over responsiveness. While this orientation aligns with many SME operational realities, it also limits adaptability under sustained structural change.

2.4 Data-Driven and Machine Learning Pricing Models

Machine learning pricing models reframe pricing as an adaptive learning problem rather than a predefined rule system. Pricing policies are derived from historical and real-time data through iterative interaction with the market.

Supervised learning approaches typically estimate demand as a function of price and contextual variables before applying optimisation procedures to generate pricing decisions (Nunan and Di Domenico, 2022). In contrast, sequential decision frameworks recognise that pricing actions influence future demand observations (Kopalle *et al.*, 2023). Reinforcement learning models extend this logic by learning pricing policies directly through interaction and addressing the exploration–exploitation trade-off (Sarkar *et al.*, 2023). Policy optimisation studies similarly prioritise long-term cumulative reward over immediate outcomes (Li and Xin, 2024).

More advanced approaches, including neural and deep learning systems, seek to capture non-linear and high-dimensional demand relationships (Rahman *et al.*, 2024). Adaptive frameworks that update continuously further assume that feedback signals are sufficiently informative to support ongoing refinement (Chen and Wang, 2023). While these approaches provide flexibility, they also depend heavily on data volume, quality, and organisational tolerance for experimentation.

Research emphasising near real-time responsiveness often equates faster adaptation with improved pricing performance (Gouveia and Mamede, 2022). However, studies on learning in dynamic markets demonstrate that non-stationarity and distributional shifts can undermine model validity even when adaptive mechanisms are present (Li and Zheng, 2024). Compared with deterministic approaches, machine learning systems prioritise responsiveness and long-term optimisation, while stability and interpretability receive comparatively less emphasis.

2.5 Data Scarcity, Uncertainty, and Practical Constraints

The literature on data scarcity and robustness challenges the assumptions underpinning many adaptive pricing systems. Rather than treating limited data as a temporary issue, these studies position it as a structural constraint affecting reliability and deployment feasibility.

Research demonstrates that predictive performance deteriorates significantly as data volume declines, particularly for flexible or high-capacity models (Peterson, 2024). Mitigation strategies such as transfer learning or data augmentation often assume access to related datasets that may not exist in SME environments. Robustness-oriented studies further critique evaluation practices that focus primarily on average-case optimisation rather than bounded degradation under adverse conditions (Zhou *et al.*, 2023).

Studies examining generalisation show that strong historical performance does not guarantee reliable deployment outcomes when distributional shifts or contextual changes occur (Chen, Simchi-Levi and Wang, 2022). Continuous updating mechanisms may also fail to resolve instability when feedback signals are sparse or delayed. Additional research highlights how pricing outputs can influence subsequent demand observations, creating feedback loops that reinforce early modelling errors over time (Jacobsen and Stewart, 2022).

Overall, this literature reframes uncertainty and limited data availability as central design concerns rather than peripheral implementation issues. For retail SMEs operating with constrained historical data, robustness may therefore be as important as optimisation performance.

2.6 Hybrid, Constrained, and Semi-Parametric Pricing Approaches

Hybrid, constrained, and semi-parametric pricing approaches attempt to balance the rigidity of rule-based systems with the flexibility of adaptive learning models. These approaches combine learning mechanisms with predefined economic or operational constraints.

Semi-parametric pricing models integrate structured economic assumptions with flexible data-driven components (Shin, Vaccari and Zeevi, 2023). Core relationships such as baseline demand or price sensitivity are specified parametrically, while residual variation is modelled through non-parametric methods. Constrained optimisation approaches similarly embed explicit limits into pricing formulations to enforce stability conditions or operational requirements (Neubert, 2022).

Rule-constrained learning frameworks restrict exploration to bounded policy spaces, limiting excessive volatility while retaining adaptive behaviour (Bansal, Sharma and Kathuria, 2022). Hybrid architectures extend this approach by combining rule systems, optimisation modules, and learning components within unified pricing frameworks (Fan, Guo and Yu, 2024).

Overall, these approaches represent a shift towards constraint-aware pricing design in which stability and interpretability become explicit objectives alongside optimisation. However, issues relating to governance, constraint evolution, and component interaction remain underdeveloped within the literature.

2.7 Pricing Stability, Fairness, and Customer-Centric Perspectives

The literature on pricing stability and fairness evaluates dynamic pricing from a customer-oriented perspective rather than solely through optimisation outcomes. In retail

environments characterised by repeated interactions, pricing behaviour may influence trust, loyalty, and perceptions of fairness.

Research on customer trust demonstrates that frequent or poorly explained price adjustments can negatively affect customer relationships even when economically justified (Kumari, Kumar and K, 2023). Studies on price fairness further show that consumers evaluate prices relative to expectations, reference points, and perceived intent (Tan *et al.*, 2025). Consequently, behavioural responses may become discontinuous and difficult to capture through standard elasticity-based models.

This literature positions price stability as an active design consideration rather than a secondary outcome of pricing systems. However, while fairness and trust are widely recognised as important, limited guidance exists regarding how these principles should be integrated into algorithmic pricing architectures.

2.8 Organisational, Ethical, and Adoption-Level Considerations

Organisational and ethical perspectives frame dynamic pricing as a socio-technical system rather than a purely computational mechanism. These studies examine how pricing systems are governed, implemented, and perceived within organisations.

Research on algorithmic pricing highlights how automation redistributes decision authority between managers and computational systems (Wang, Liu and Yang, 2023). Delegating pricing decisions to opaque models raises concerns regarding accountability, oversight, and managerial control, particularly in SMEs with limited analytical capacity.

Studies linking pricing systems with broader digital transformation processes emphasise the importance of data infrastructure and organisational readiness (Jullien, Reisinger and Rey, 2023). Adoption-focused research further suggests that implementation decisions are shaped by perceived risk, uncertainty, and concern regarding customer reaction (Cohen, Miao and Wang, 2025). Pricing automation is therefore often introduced incrementally rather than through radical organisational transformation.

Overall, these perspectives demonstrate that pricing systems cannot be evaluated solely through optimisation performance. Governance structures, reputational concerns, and operational readiness significantly influence implementation feasibility and long-term acceptance.

2.9 Summary

This chapter has critically reviewed the dynamic pricing literature across conceptual, methodological, behavioural, and organisational perspectives. The review demonstrates clear differences between deterministic, machine learning-based, and hybrid pricing approaches. Deterministic systems prioritise transparency, stability, and managerial control, whereas machine learning approaches emphasise adaptivity and long-term optimisation. Hybrid models attempt to balance these competing objectives by combining learning mechanisms with operational constraints. However, much of the literature continues to rely on assumptions regarding stable demand conditions, sufficient data availability, and organisational readiness that may not reflect the realities of retail SMEs.

The review also highlights that customer-oriented and organisational considerations, including pricing stability, fairness, governance, and implementation feasibility, are frequently treated as secondary concerns within pricing system evaluation. Consequently, limited research

comparatively evaluates pricing paradigms within constrained SME retail environments using criteria that extend beyond optimisation performance alone. This gap motivates the comparative framework adopted in this dissertation, which evaluates rule-based, machine learning-based, and hybrid pricing models in terms of both revenue performance and pricing stability under consistent retail SME conditions.

Chapter 3: Methodology

3.1 Introduction

This chapter presents the methodological framework used to evaluate rule-based, machine learning-based, and hybrid dynamic pricing models within a retail SME context. The methodology supports a structured comparison of these approaches under consistent experimental conditions and aligns with the study's aim of comparative evaluation rather than the development of new pricing algorithms. Revenue performance and pricing stability are incorporated as the primary evaluation dimensions.

The chapter outlines the research design, research approach, dataset selection, data preparation procedures, model implementation, execution pipeline, evaluation metrics, and ethical considerations. The methodological choices reflect both theoretical considerations and practical constraints associated with retail SME environments.

3.2 Research Design

This study adopts a quantitative, computational research design based on **offline simulation** using historical retail data. Pricing strategies are evaluated under controlled experimental conditions rather than through live commercial deployment. This enables systematic comparison of model behaviour without affecting actual customers or introducing operational risks.

A comparative framework is used in which all models operate on the same dataset and are evaluated using identical performance metrics. By maintaining a shared analytical structure, differences in outcomes can be attributed directly to pricing logic rather than inconsistencies in

data processing or evaluation procedures. Offline simulation is particularly appropriate for pricing research because real-world experimentation may introduce financial and ethical risks associated with visible price fluctuations.

Overall, the research design ensures consistency, comparability, and practical feasibility within a controlled experimental environment.

3.3 Research Approach

This study adopts a comparative, model-based research approach to evaluate rule-based, machine learning-based, and hybrid pricing strategies. The focus is on analysing how these approaches perform under identical experimental conditions rather than proposing a new pricing framework.

The selected models represent distinct pricing paradigms. Rule-based pricing relies on predefined heuristics that provide stable and interpretable behaviour. Machine learning-based pricing uses demand estimation to adapt pricing decisions dynamically. The hybrid model combines optimisation with constraints designed to reduce excessive price variation, thereby balancing responsiveness and control.

All models are implemented within a unified simulation framework using the same dataset and evaluation metrics. This ensures consistent comparison across pricing strategies and supports direct evaluation of revenue performance and pricing stability.

3.4 Dataset Description

This study uses the **Online Retail II dataset** obtained from a publicly available Kaggle repository, focusing on transactional records from 2010–2011 (Kabilan, no date). The dataset

contains transaction-level information including product identifiers, quantities, prices, and timestamps, enabling analysis of pricing behaviour in relation to observed demand patterns.

Although the dataset includes transactions from multiple countries, the analysis is restricted to the United Kingdom in order to maintain a consistent market environment and reduce variability associated with cross-country economic conditions and consumer behaviour.

To support pricing simulation, the dataset is aggregated into a daily product-level structure containing units sold, average price, and revenue for each product. The analysis focuses on products with sustained activity and observable price variation to ensure meaningful demand and pricing dynamics.

Dataset Summary Statistics

Metric	Value
Original Transaction Records	541,910
Valid Transaction Records	484,082
Unique Products	3,801
Selected Products	5
Aggregated Daily Observations	1,358
Final Modelling Observations	1,323
Training Observations	1,057
Testing Observations	266

Table 3.1: Dataset Summary Statistics

The filtering and aggregation process reduced the size of the original transactional dataset while preserving products with sufficient temporal and behavioural variation for comparative evaluation.

Selected Products for Comparative Evaluation

Stock Code	Product Description	Active Days
22423	Regency Cakestand 3 Tier	301
85123A	White Hanging Heart T-Light Holder	305
47566	Party Bunting	291
85099B	Jumbo Bag Red Retrospot	300
22086	Paper Chain Kit 50's Christmas	161

n = 5 products

Note: Active Days refers to the number of days with recorded transactions within the study period.

Table 3.2: Selected Products for Comparative Evaluation

The selected products represent items with sustained transaction activity and measurable pricing variation throughout the evaluation period. Using products with differing behavioural characteristics strengthens the robustness of the comparative analysis.

3.5 Data Preparation

The data preparation process transforms raw transactional records into a structured dataset suitable for demand modelling and pricing simulation. The process includes data cleaning, aggregation, feature engineering, and chronological dataset partitioning.

Transactions that do not represent valid retail demand are removed during the cleaning stage. Cancelled invoices, negative quantities associated with returns, and records containing zero or negative prices are excluded to prevent distortion of demand measurements. Non-product stock

codes and extreme price outliers are also removed to improve data quality and maintain realistic pricing conditions.

Following cleaning, the dataset is aggregated into a daily product-level structure. Total units sold represent daily demand, average price captures prevailing price levels, and daily revenue is calculated as the product of price and quantity. Feature engineering is then applied to capture temporal demand behaviour. Lagged demand variables, rolling demand averages, and calendar-based features such as day-of-week and month indicators are included to support demand estimation.

The dataset is divided into training and testing subsets using a chronological split. Earlier observations are used for model training, while later observations are reserved for evaluation. This preserves temporal ordering and prevents information leakage.

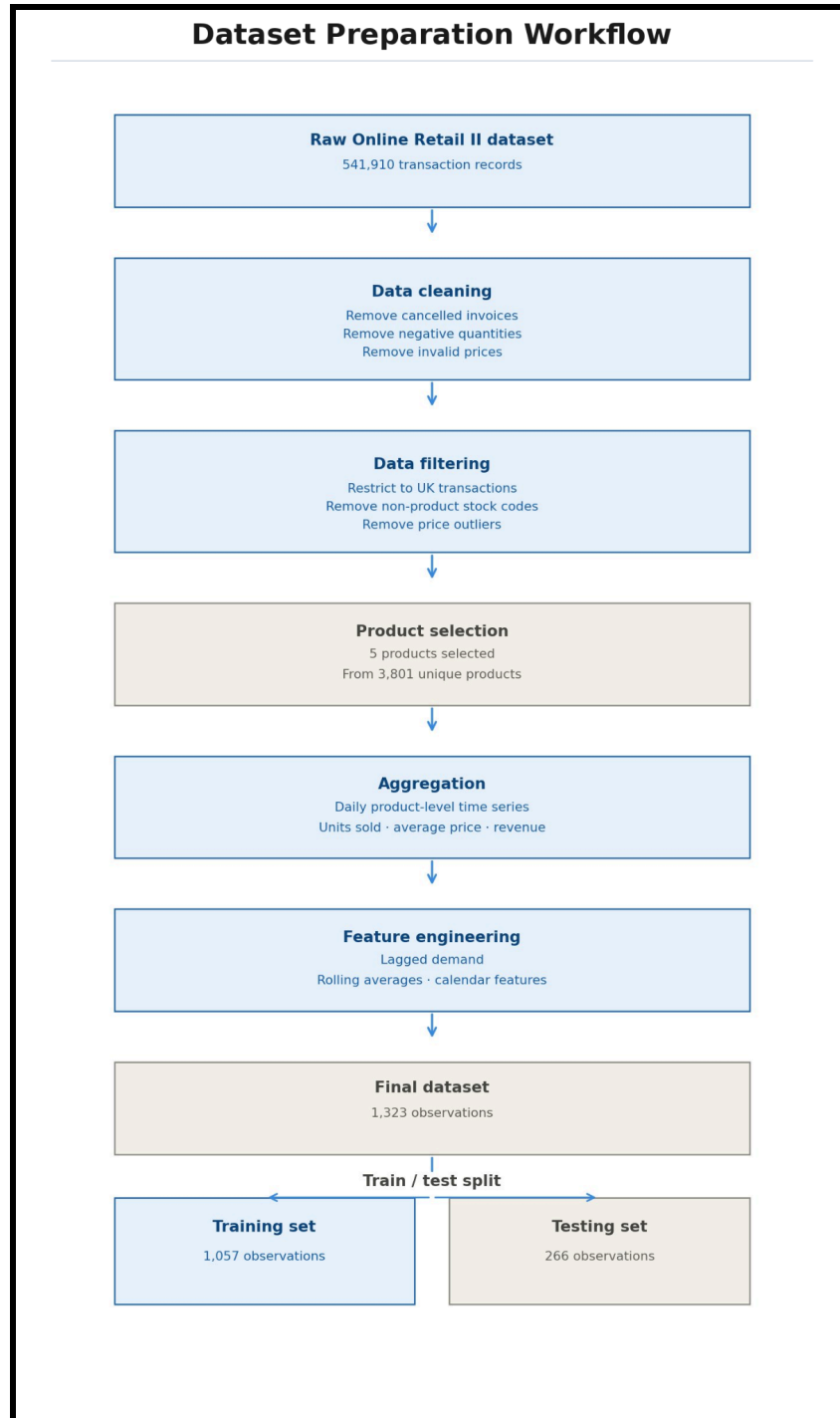


Figure 3.1: Dataset Dataset Preparation Workflow

Figure 3.1 illustrates the sequence of cleaning, filtering, aggregation, and feature engineering stages used to construct the final analytical dataset from the original transactional records.

3.6 Model Implementation

This section outlines the implementation of the three pricing strategies evaluated in the study. All models operate within a shared simulation framework using identical inputs and candidate price sets to ensure consistent comparison.

3.6.1 Rule-Based Model

The rule-based model applies a deterministic pricing strategy based on predefined percentage adjustments. Predicted demand at the current price is compared with a rolling average of recent demand observations. If predicted demand exceeds the rolling average, price is increased by 2%; if demand falls below the average, price is reduced by 2%. Prices remain unchanged when demand is stable.

The resulting price is mapped to the nearest option within the candidate price set. This approach produces stable and interpretable pricing behaviour but offers limited responsiveness to rapidly changing demand conditions.

3.6.2 Machine Learning Model

The machine learning model uses a linear regression framework trained on lagged demand variables, price, and calendar-based features. For each observation, five candidate prices are generated within a $\pm 5\%$ range around the base price. Demand is predicted for each candidate price, and expected revenue is calculated as the product of predicted demand and price.

The candidate price that maximises predicted revenue is selected. This enables adaptive pricing behaviour under changing demand conditions. However, because optimisation is unconstrained, the model may produce higher pricing volatility.

3.6.3 Hybrid Model

The hybrid model combines optimisation with stability-oriented constraints. It first identifies the revenue-maximising price using the machine learning approach before applying constraints that limit the magnitude of price changes relative to the previous simulation step.

A weighted smoothing mechanism moderates transitions between consecutive prices, reducing abrupt fluctuations and improving temporal consistency. The final price is then mapped to the nearest candidate price within the predefined price set. This design **balances adaptability with controlled price movement**, making the model particularly relevant for SME pricing contexts.

3.7 Execution Pipeline and System Workflow

The study is implemented as a structured computational pipeline consisting of three stages: data preparation and model training, pricing simulation, and evaluation. The pipeline supports both integrated and staged execution while maintaining deterministic processing throughout.

In integrated execution, preprocessing, feature construction, demand model training, pricing simulation, and evaluation are performed sequentially within a single workflow. This approach is used to generate the final results presented in the study and ensures that all pricing strategies are evaluated under identical conditions.

The pipeline may also be executed in separate stages. The first stage performs data preparation and demand model training. The second stage generates pricing decisions for the rule-based, machine learning-based, and hybrid strategies. The final stage computes evaluation metrics and statistical test results.

This staged structure improves methodological transparency by allowing intermediate outputs to be inspected without altering subsequent processing stages. The use of fixed data inputs and deterministic execution ensures reproducibility across repeated runs.

An interactive visualisation interface is also implemented to present evaluation outputs including revenue metrics, stability measures, comparative summaries, and statistical testing results. The interface functions independently from the simulation pipeline and does not influence experimental outcomes.

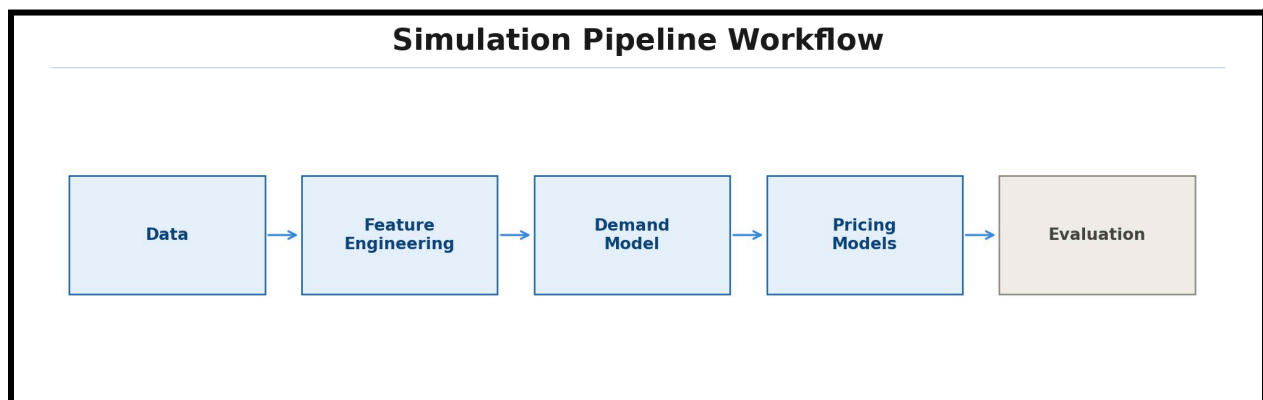
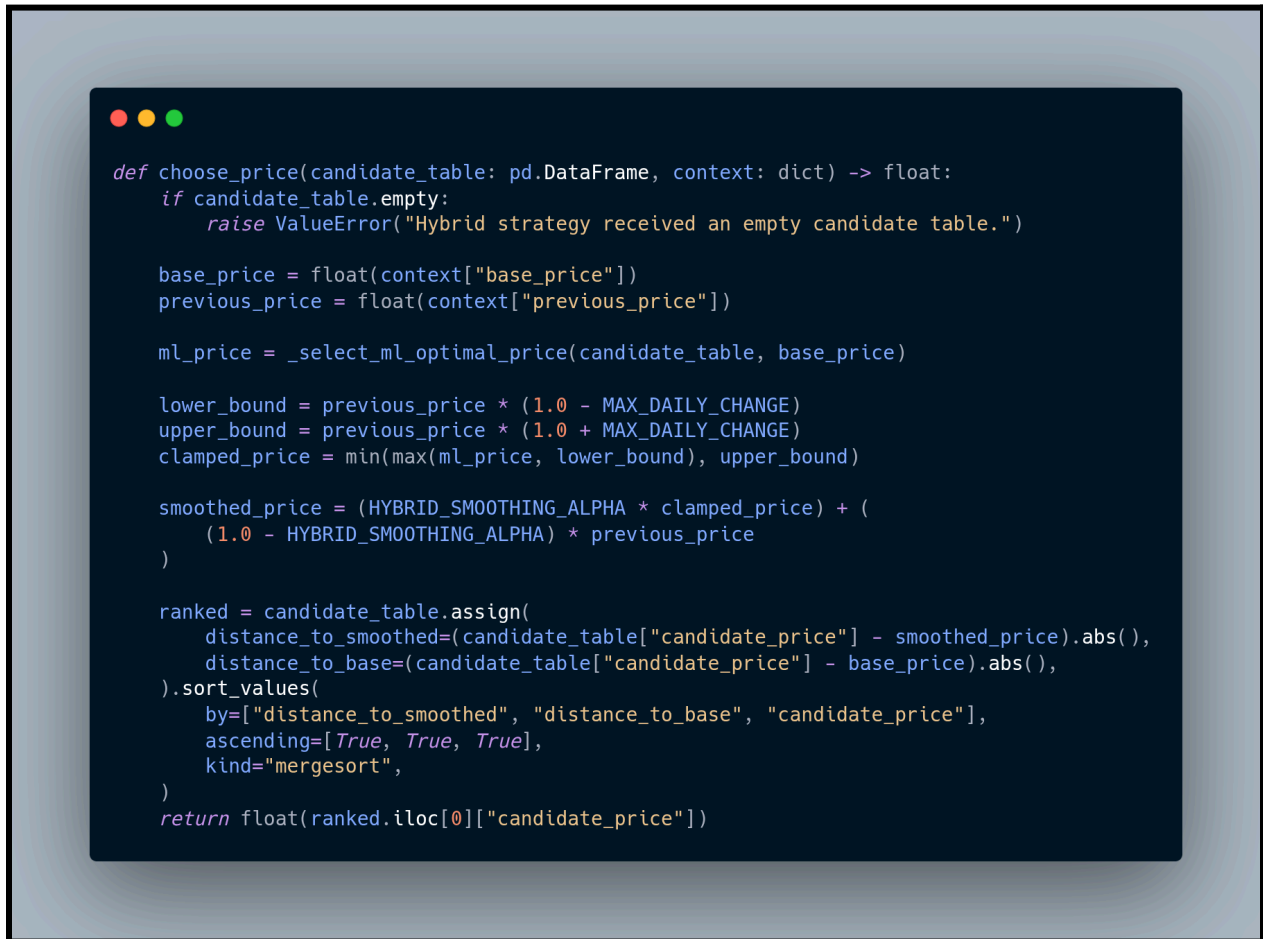


Figure 3.2: Simulation pipeline for dynamic pricing model evaluation

Figure 3.2 illustrates the overall simulation workflow used throughout the study, including preprocessing, model training, pricing simulation, and evaluation stages executed within the deterministic analytical framework.

3.7.1 Implementation Illustration



```
def choose_price(candidate_table: pd.DataFrame, context: dict) -> float:
    if candidate_table.empty:
        raise ValueError("Hybrid strategy received an empty candidate table.")

    base_price = float(context["base_price"])
    previous_price = float(context["previous_price"])

    ml_price = _select_ml_optimal_price(candidate_table, base_price)

    lower_bound = previous_price * (1.0 - MAX_DAILY_CHANGE)
    upper_bound = previous_price * (1.0 + MAX_DAILY_CHANGE)
    clamped_price = min(max(ml_price, lower_bound), upper_bound)

    smoothed_price = (HYBRID_SMOOTHING_ALPHA * clamped_price) + (
        (1.0 - HYBRID_SMOOTHING_ALPHA) * previous_price
    )

    ranked = candidate_table.assign(
        distance_to_smoothed=(candidate_table["candidate_price"] - smoothed_price).abs(),
        distance_to_base=(candidate_table["candidate_price"] - base_price).abs(),
    ).sort_values(
        by=["distance_to_smoothed", "distance_to_base", "candidate_price"],
        ascending=[True, True, True],
        kind="mergesort",
    )
    return float(ranked.iloc[0]["candidate_price"])
```

Figure 3.3: Hybrid pricing function implementing optimisation, constraint, and smoothing mechanisms

The implementation excerpt shown in Figure 3.3 demonstrates how the hybrid pricing strategy combines optimisation with constraint-based control and smoothing mechanisms. The function first identifies the revenue-maximising candidate price before limiting excessive variation relative to previous prices. A smoothing step is then applied to moderate transitions between observations and reduce volatility.

The complete implementation of the pricing engine, simulation pipeline, and evaluation framework is publicly available to support transparency and reproducibility (Gore, 2026).

3.8 Evaluation Metrics

This study evaluates pricing strategies using two primary dimensions: **revenue performance** and **pricing stability**. Revenue performance is assessed using total revenue and mean daily revenue, while pricing stability is evaluated using mean absolute price change, price standard deviation, maximum price jump, and change frequency.

The use of multiple stability metrics ensures that both the magnitude and frequency of price variation are captured. This is particularly relevant in SME retail environments where excessive volatility may influence customer perception and operational consistency.

Metrics are computed at both product and strategy levels to support detailed comparative analysis. Statistical testing, including paired t-tests and Wilcoxon tests, is used to assess whether observed differences are statistically significant.

3.9 Ethical Considerations

This study uses publicly available transactional data and does not involve direct interaction with human participants. The analysis is conducted using aggregated product-level data, ensuring that no personally identifiable information is processed.

The research is based on offline simulation and does not involve real-world deployment of pricing strategies. Consequently, no customers are affected by experimental pricing decisions. The study nevertheless acknowledges ethical concerns associated with dynamic pricing,

particularly the potential effects of excessive price variation on customer trust and perceptions of fairness.

The inclusion of pricing stability as an evaluation dimension reflects these concerns and supports a more balanced assessment of pricing behaviour. Transparency and methodological consistency are maintained by applying all pricing models under identical experimental conditions.

3.10 Summary

This chapter has presented the methodological framework used to evaluate dynamic pricing strategies within a retail SME context. It has outlined the research design, dataset preparation process, model implementation, execution pipeline, evaluation metrics, and ethical considerations supporting the comparative framework.

By applying all pricing strategies under identical conditions, the methodology ensures that differences in outcomes can be attributed directly to pricing logic rather than inconsistencies in experimental structure. The framework therefore provides a reproducible basis for evaluating revenue performance and pricing stability across alternative pricing paradigms. The following chapter presents the results derived from this methodology.

Chapter 4: Results

4.1 Introduction

This chapter presents the empirical results obtained from the comparative evaluation of rule-based, machine learning-based, and hybrid dynamic pricing models. The results are generated using the experimental framework described in Chapter 3, in which all models were evaluated under identical conditions using a shared dataset and simulation environment. This ensures that observed differences in outcomes are attributable to pricing logic rather than inconsistencies in data handling or evaluation procedures.

The chapter reports findings across the two primary evaluation dimensions: revenue performance and pricing stability. The analysis begins with dataset outcomes before progressing to individual model results, comparative evaluation, and statistical testing. The presentation remains descriptive, with interpretation reserved for Chapter 5.

4.2 Data Processing Results

The final dataset consisted of 484,082 valid transaction records derived from an initial set of 541,910 observations after removing cancelled invoices, returns, and invalid pricing entries. To maintain a consistent market environment, the analysis was restricted to transactions originating from the United Kingdom.

A subset of five products was selected from 3,801 unique items based on sustained activity, revenue contribution, and observable price variation. The dataset was aggregated into a daily product-level structure containing 1,358 observations representing units sold, average price, and

revenue over time. Following feature construction and filtering procedures, 1,323 observations remained for modelling and evaluation.

The dataset was divided chronologically into 1,057 training observations and 266 testing observations to preserve temporal ordering and prevent information leakage. This structure ensured that all pricing models operated under identical testing conditions and comparable demand histories.

4.3 Model Results

4.3.1 Rule-Based Model Results

The rule-based pricing model generated a total simulated revenue of 384,707.66 and a mean daily revenue of 1,436.04. The deterministic structure of the model produced predictable and operationally interpretable pricing behaviour throughout the evaluation period.

In terms of stability, the model recorded a mean absolute price change of 1.1273 and a price standard deviation of 1.0127, indicating moderate pricing variability over time. The maximum observed price jump reached 8.6071, while the change frequency of 0.9514 showed that prices were updated regularly throughout the simulation.

At the product level, higher-demand products contributed a greater proportion of overall revenue, although the model maintained structurally consistent pricing behaviour across all evaluated items. Overall, the rule-based approach produced **moderate outcomes across both revenue and pricing stability dimensions**.

4.3.2 Machine Learning Model Results

The machine learning pricing model achieved the **highest revenue performance** among all evaluated strategies. The model generated a total simulated revenue of 393,990.38 and a mean daily revenue of 1,470.96 by selecting prices that maximised predicted revenue for each observation.

However, the model also demonstrated the **highest pricing variability**. The mean absolute price change reached 1.1388, while the price standard deviation of 1.0346 reflected broader variation in pricing levels. The maximum price jump of 8.8170 represented the highest extreme movement among all strategies. Despite this variability, the change frequency remained comparable to the other models at 0.9480.

At the product level, the machine learning model consistently generated higher revenue outcomes across all evaluated products. These findings indicate strong optimisation capability but increased price volatility relative to the alternative approaches.

4.3.3 Hybrid Model Results

The hybrid pricing model produced balanced outcomes by combining optimisation with constraint-based control mechanisms. The model achieved a total simulated revenue of 378,588.42 and a mean daily revenue of 1,413.63, placing its financial performance below the machine learning and rule-based strategies but within a comparable overall range.

The hybrid model achieved the **strongest pricing stability outcomes** among all evaluated strategies. The mean absolute price change was 0.7600, while the price standard deviation of 0.8027 reflected comparatively consistent pricing levels over time. The maximum observed price

jump of 6.7280 remained lower than those recorded by the other models. The change frequency of 0.9246 further showed that the model maintained regular updates while moderating adjustment magnitude.

At the product level, the hybrid strategy produced smoother pricing transitions and reduced cumulative volatility across all evaluated products. These findings demonstrate that adaptive pricing behaviour can be moderated without substantial reductions in revenue performance.

4.4 Comparative Analysis

This section compares the performance of the three pricing strategies across revenue and stability metrics using a shared evaluation framework.

Strategy-Level Performance Metrics						
Strategy	Revenue Metrics		Pricing Stability Metrics			
	Total Revenue (£)	Mean Daily Revenue (£)	Mean Abs. Price Change	Price Std Dev	Max Price Jump	Change Frequency
Rule-Based	£384,707.66	£1,436.04	1.1273	1.0127	8.6071	0.9514
Machine Learning	£393,990.38	£1,470.96	1.1388	1.0346	8.8170	0.9480
Hybrid	£378,588.42	£1,413.63	0.7600	0.8027	6.7280	0.9246

Note: Bold green values indicate best performance per metric. Revenue metrics: higher is better. Stability metrics: lower is better.

Table 4.1: Strategy-Level Performance Metrics

Table 4.1 summarises the aggregated performance outcomes across all pricing strategies. The machine learning model achieved the highest total and mean revenue values, followed by the rule-based model, while the hybrid model recorded the lowest revenue performance. However, the overall differences between strategies remained moderate.

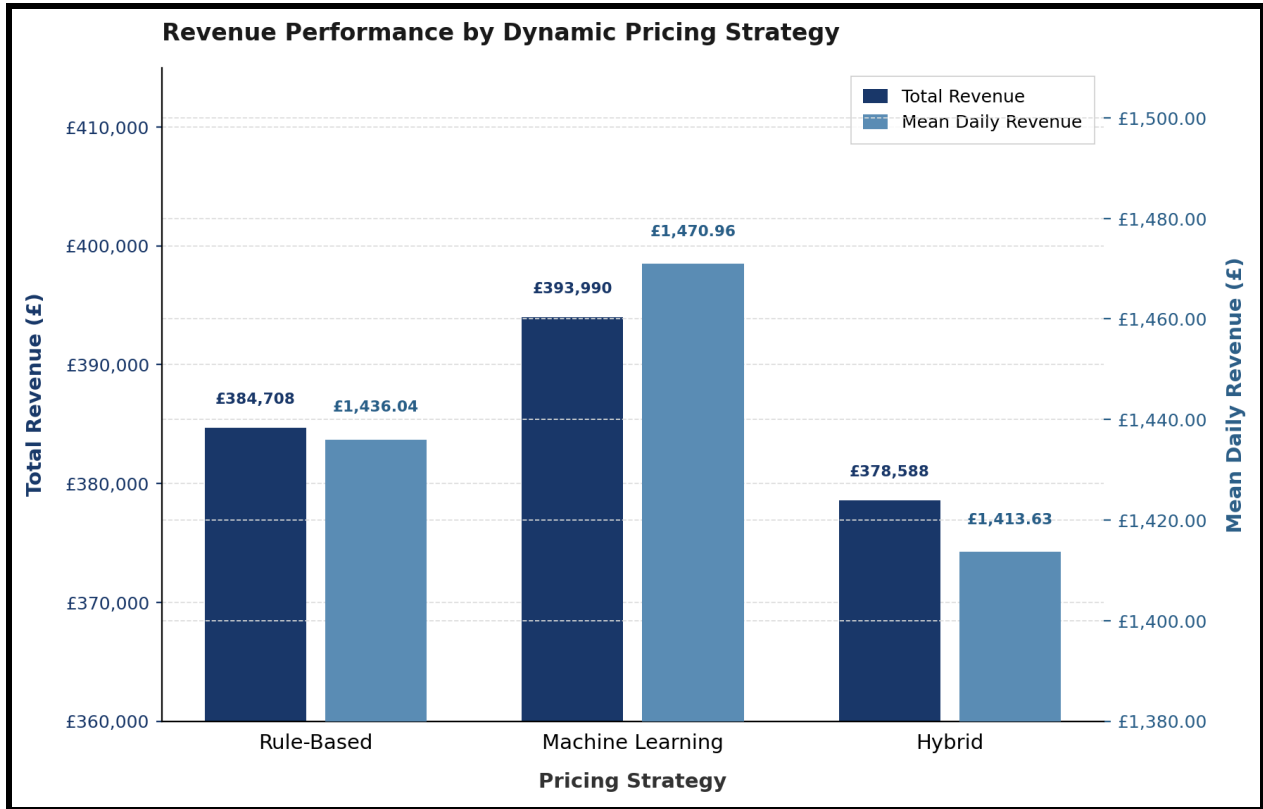


Figure 4.1: Comparative Revenue Performance Across Pricing Models

Figure 4.1 visually compares total and mean daily revenue across the evaluated pricing strategies. The machine learning model achieved the strongest financial performance, while the hybrid model produced comparatively lower revenue outcomes.

In contrast, pricing stability metrics showed a different ranking pattern. The hybrid model consistently achieved the lowest levels of price variation across all stability measures, while the machine learning and rule-based models demonstrated higher variability.

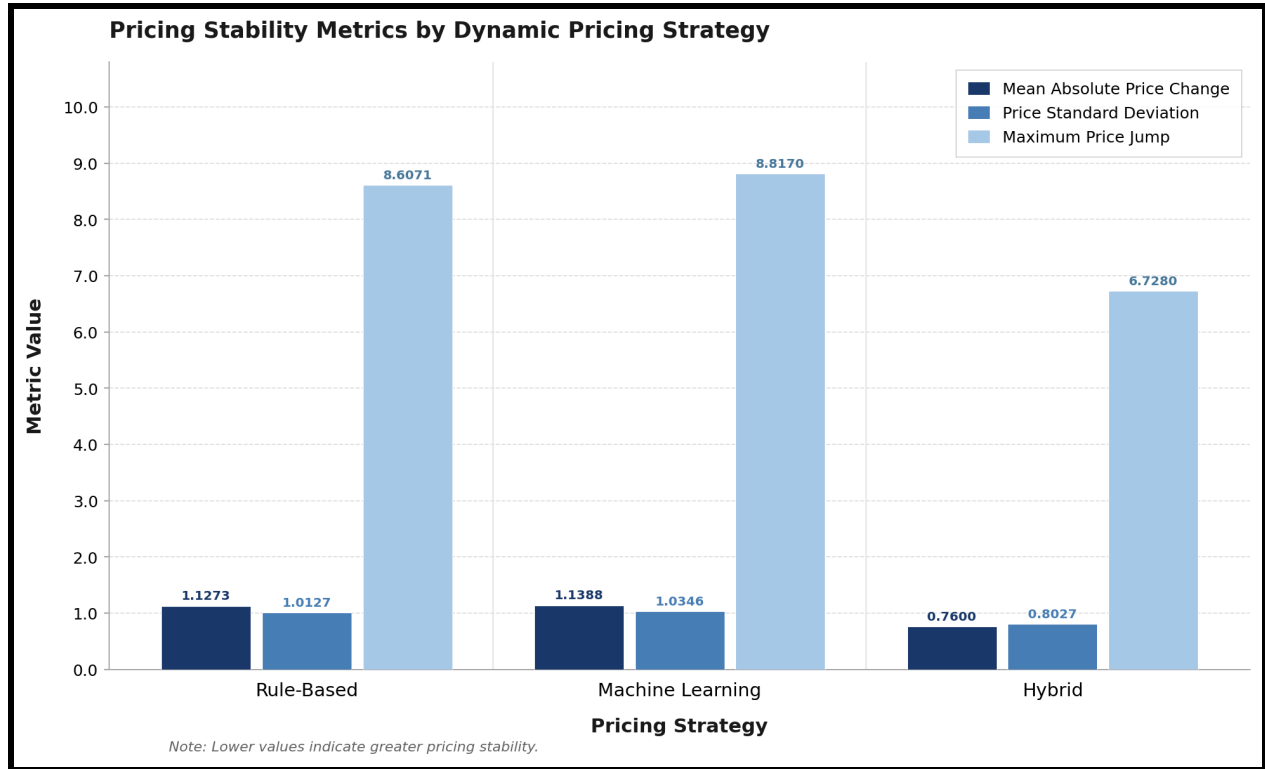


Figure 4.2: Comparative Pricing Stability Metrics Across Pricing Models

Figure 4.2 presents the comparative pricing stability metrics across the evaluated strategies. The hybrid model consistently recorded lower values for mean absolute price change, price standard deviation, and maximum price jump, indicating smoother pricing behaviour over time.

The machine learning model recorded the highest maximum price jump, followed closely by the rule-based strategy. Although price change frequency remained high across all models, the key distinction was the magnitude rather than the frequency of adjustments. The hybrid model maintained regular updates while moderating price fluctuations.

Overall, the comparative analysis demonstrates a **clear trade-off between optimisation and pricing stability**. The machine learning model achieved the strongest revenue performance but

with higher volatility, whereas the hybrid model prioritised stability with slightly lower revenue outcomes. The rule-based model occupied an intermediate position across both dimensions.

4.4.1 Product-Level Comparative Results

Product-level analysis was conducted to examine whether the comparative performance patterns remained consistent across individual stock codes.

Across all evaluated products, the machine learning model consistently generated the highest simulated revenue outcomes. Products with stronger baseline demand, particularly stock code 22423, produced the largest differences in revenue performance between pricing strategies. Although the hybrid model generally recorded lower revenue values, these differences remained moderate relative to the overall revenue scale.

The stability metrics showed a different ordering. The hybrid model consistently achieved lower values for mean absolute price change, price standard deviation, and maximum price jump across all products. This reduction in variability was particularly visible for products characterised by stronger fluctuations in predicted demand conditions.

Overall, the product-level analysis confirmed that the comparative differences between pricing strategies were systematic rather than isolated to specific products. The machine learning model consistently prioritised revenue optimisation, while the hybrid model consistently prioritised pricing stability.

4.5 Statistical Testing

This section presents the statistical tests used to evaluate whether the observed differences between pricing models were **statistically significant**. Paired t-tests and Wilcoxon signed-rank tests were conducted using aligned observations across all pricing strategies.

Statistical Testing Results					
	Comparison	Metric	Statistical Test	Result	Significant at $p < 0.05$
Revenue	Hybrid vs Machine Learning	Revenue	Paired t-test	Significant	Yes
			Wilcoxon signed-rank	Significant	Yes
	Hybrid vs Rule-Based	Revenue	Paired t-test	Significant	Yes
			Wilcoxon signed-rank	Not Significant	No
Stability	Hybrid vs Machine Learning	Stability	Paired t-test	Significant	Yes
			Wilcoxon signed-rank	Significant	Yes
	Hybrid vs Rule-Based	Stability	Paired t-test	Significant	Yes
			Wilcoxon signed-rank	Significant	Yes

Note: Significance assessed at $\alpha = 0.05$. Dashed line separates revenue and stability metric groups. *Italic test names follow standard statistical nomenclature.*

Table 4.2: Statistical Testing Results

Table 4.2 summarises the statistical testing outcomes across the evaluated pricing strategies. The results indicate significant differences between the hybrid and machine learning models across both revenue and pricing stability dimensions.

For revenue performance, comparisons between the hybrid and machine learning models produced statistically significant results under both testing methods, with p-values below the

0.05 threshold. These findings support the comparative analysis in which the machine learning model achieved higher revenue performance than the hybrid strategy.

Comparisons between the hybrid and rule-based models produced less consistent results. The paired t-test indicated statistical significance, whereas the Wilcoxon signed-rank test did not. This suggests that the revenue difference between these models may be sensitive to distributional characteristics within the evaluation data.

For pricing stability, both statistical tests consistently identified significant differences between the hybrid model and the alternative strategies. The resulting p-values remained below the 0.05 threshold across all stability-related comparisons, confirming that the hybrid model produced lower levels of price variation throughout the evaluation period.

The statistical analysis was conducted using 266 paired observations derived from the testing dataset. This aligned structure ensured that all comparisons were performed under identical evaluation conditions, strengthening the validity of the comparative findings.

4.6 Summary

This chapter has presented the results of the comparative evaluation of rule-based, machine learning-based, and hybrid pricing models. The analysis included dataset characteristics, individual model performance, comparative evaluation, and statistical testing within a shared experimental framework.

The machine learning model achieved the highest revenue performance across all evaluated metrics, while the hybrid model demonstrated the strongest pricing stability outcomes. The

rule-based model produced intermediate results across both dimensions. These patterns remained consistent across both aggregate and product-level analysis.

Statistical testing confirmed significant differences in pricing stability between the hybrid model and the alternative strategies, while revenue differences between the hybrid and rule-based models were less consistent. Together, these findings provide the empirical foundation for the discussion presented in the following chapter.

Chapter 5: Discussion

5.1 Introduction

This chapter interprets the findings presented in Chapter 4 in relation to the research question and the literature on dynamic pricing. While the previous chapter focused on reporting empirical outcomes, this chapter examines how and why the observed differences between pricing models emerged.

The discussion focuses on the two primary evaluation dimensions established in this study: revenue performance and pricing stability. These dimensions reflect the central trade-off underlying dynamic pricing systems, particularly within retail SME contexts. The findings are interpreted in relation to theoretical perspectives presented in Chapter 2 and evaluated in terms of their practical relevance for SME pricing decisions.

The chapter is organised into four sections. Section 5.2 interprets the results, Section 5.3 links the findings to existing literature, Section 5.4 discusses practical implications, and Section 5.5 outlines the study limitations before the concluding summary.

5.2 Interpretation of Results

The findings reveal clear differences in how the three pricing strategies balance revenue performance and pricing stability. These differences emerge from the structural characteristics of each model rather than incidental variation within the dataset.

The machine learning model achieved the highest revenue performance across all evaluated metrics, reflecting its optimisation-driven design in which candidate prices are selected

according to expected revenue outcomes. However, the model also produced the highest levels of pricing variability. Metrics such as mean absolute price change, price standard deviation, and maximum price jump demonstrate that the model responds aggressively to estimated demand conditions without mechanisms to moderate price movement.

The rule-based model produced moderate outcomes across both revenue and stability dimensions. Its deterministic pricing logic generated predictable and interpretable behaviour, although cumulative variability remained comparatively high because prices were updated frequently across consecutive periods. This suggests that predefined heuristics alone may not sufficiently regulate pricing volatility over time.

The **hybrid model achieved the strongest pricing stability outcomes** by combining optimisation with bounded adjustment constraints and smoothing mechanisms. By limiting the magnitude of price changes relative to previous prices, the model generated smoother pricing transitions and reduced abrupt fluctuations throughout the evaluation period. The stateful structure of the model further contributed to greater temporal consistency.

Although the hybrid strategy achieved lower revenue performance than the machine learning and rule-based approaches, the reduction remained moderate relative to the improvement in stability. This suggests that **optimisation and pricing consistency should not necessarily be treated as mutually exclusive objectives**. Instead, the findings indicate that constraint-aware pricing systems can balance adaptability with controlled pricing behaviour.

An additional insight is that the primary distinction between pricing strategies lies in the magnitude rather than the frequency of price adjustments. All models updated prices regularly throughout the evaluation period, but the hybrid approach moderated the scale of adjustments

and therefore produced smoother pricing behaviour. This may be particularly important in retail environments where customers are sensitive to abrupt or excessive price fluctuations.

Overall, the results demonstrate that machine learning approaches prioritise optimisation and responsiveness, rule-based systems emphasise deterministic control, and hybrid models integrate adaptive pricing with explicit stability mechanisms. These structural differences directly shaped the observed outcomes across both revenue and pricing stability dimensions.

5.3 Link to Literature

The findings align closely with themes identified in the dynamic pricing literature, particularly regarding the trade-off between optimisation and pricing stability. Machine learning pricing models are widely characterised as adaptive and optimisation-driven systems that prioritise revenue maximisation through data-driven decision-making (Kopalle *et al.*, 2023). The results support this perspective, as the machine learning model achieved the highest revenue while also exhibiting the greatest pricing variability.

The behaviour of the rule-based model is similarly consistent with literature describing deterministic pricing systems as stable, interpretable, and operationally simple, but less responsive to changing demand conditions (Das *et al.*, 2024). Although the model generated predictable pricing behaviour, the relatively high frequency of adjustments indicates that deterministic heuristics alone may not sufficiently control cumulative volatility.

The hybrid model provides empirical support for research on constrained and semi-adaptive pricing approaches. Hybrid systems are designed to combine optimisation with explicit constraints on price movement (Shin, Vaccari and Zeevi, 2023). The results demonstrate that this

design can significantly reduce volatility while maintaining competitive revenue performance, supporting the argument that constraint-aware systems are particularly suitable for uncertain or data-constrained environments.

The findings also reinforce literature addressing data limitations within SME settings. Many adaptive pricing models assume large volumes of reliable data and continuous feedback for refinement (Gouveia and Mamede, 2022). In practice, SMEs often operate with limited analytical infrastructure, which may increase prediction instability and contribute to pricing volatility within machine learning systems.

Customer-oriented perspectives on pricing stability and fairness are also reflected in the results. Previous research suggests that abrupt or frequent price changes may negatively affect customer trust and perceived fairness (Diao, Harutyunyan and Jiang, 2023). Although customer perception was not directly measured in this study, the stability metrics provide an indirect representation of pricing consistency. The smoother behaviour observed in the hybrid model therefore aligns with recommendations for more controlled pricing practices.

Finally, the study contributes to a gap identified in the literature regarding comparative evaluation of pricing models within SME-focused contexts using both revenue and stability metrics. By applying all strategies within a consistent experimental framework, the research extends existing optimisation-focused studies and provides a more balanced assessment of pricing system performance.

5.4 Practical Implications

The findings have important implications for pricing decisions within retail SMEs. Although dynamic pricing systems are frequently evaluated primarily according to revenue generation, the results demonstrate that pricing stability is also a critical consideration within customer-facing retail environments.

The machine learning model achieved the strongest revenue performance but also generated the highest levels of price volatility. In practical SME settings, such behaviour may be difficult to manage operationally and could negatively influence customer perceptions of fairness and consistency. In addition, machine learning systems require reliable data, monitoring capability, and analytical infrastructure, which may not always be available within smaller organisations.

The hybrid model represents a more balanced alternative for SMEs seeking to combine adaptive pricing behaviour with operational control. By incorporating bounded adjustments and smoothing mechanisms, the model significantly reduced pricing variability while maintaining comparatively competitive revenue outcomes. This makes the approach particularly suitable for retail environments where pricing consistency and customer trust are important considerations.

The rule-based model remains relevant for SMEs with limited technical capability or restricted historical data. Its deterministic structure makes pricing decisions transparent and relatively easy to implement. However, the findings suggest that purely rule-based systems may struggle to fully optimise revenue performance while also failing to adequately regulate cumulative variability.

An important practical insight is that the **trade-off between optimisation and stability is not absolute**. The findings demonstrate that substantial improvements in pricing consistency can be

achieved with only moderate reductions in revenue performance. This suggests that SMEs do not necessarily need to prioritise one objective entirely at the expense of the other.

The results also highlight the importance of evaluating pricing systems using multiple performance dimensions rather than relying solely on revenue metrics. Stability indicators such as price dispersion, adjustment magnitude, and maximum price change provide additional insight into how pricing behaviour evolves over time. Incorporating these measures enables SMEs to select pricing approaches that align more effectively with operational priorities and customer expectations.

5.5 Limitations

This study has several limitations that should be considered when interpreting the findings. First, the analysis is based on a single retail dataset restricted to the United Kingdom and a subset of five products. While this supports consistency within the experimental framework, it limits the generalisability of the findings to other retail markets or product categories.

Second, the study relies on a linear regression model for demand estimation. Although this ensures methodological consistency across pricing strategies, the model may not fully capture complex or non-linear demand relationships. The results should therefore be interpreted as comparative rather than absolute indicators of pricing performance.

Third, the research is based on offline simulation rather than real-world deployment. Pricing decisions are evaluated using predicted demand rather than actual customer responses, meaning that behavioural effects such as fairness perception, reference price formation, or long-term purchasing adaptation are not directly captured.

The study also adopts a simplified representation of pricing environments. Factors such as competitor pricing, inventory constraints, promotional strategies, and supply chain dynamics are excluded in order to maintain a controlled comparative framework. While this isolates the impact of pricing logic, it limits the external validity of the findings.

Finally, the evaluation framework focuses primarily on quantitative measures related to revenue and pricing stability. Qualitative dimensions such as managerial usability and direct customer perception were not examined. Future research could incorporate behavioural and qualitative evaluation measures to provide a more comprehensive assessment of pricing systems.

5.6 Summary

This chapter has interpreted the comparative performance of rule-based, machine learning-based, and hybrid pricing models within a retail SME context. The findings demonstrate that the machine learning model maximised revenue performance but introduced greater pricing variability, whereas the hybrid model achieved stronger pricing stability with only moderate reductions in revenue outcomes. The rule-based model produced intermediate behaviour across both dimensions.

The discussion linked these findings to existing literature on optimisation-driven, deterministic, and constraint-aware pricing systems, while also highlighting their practical implications for SME pricing decisions. The analysis suggests that effective pricing systems should be evaluated using both revenue and stability measures rather than optimisation outcomes alone.

Although the study is subject to limitations relating to data scope, modelling simplicity, and simulation-based evaluation, it provides a structured and reproducible framework for

comparative pricing analysis. Overall, the findings support a more balanced approach to dynamic pricing evaluation in retail SME environments, where operational consistency and customer-oriented pricing behaviour remain important considerations.

Chapter 6: Conclusion

6.1 Conclusion

This dissertation comparatively evaluated rule-based, machine learning-based, and hybrid dynamic pricing models within a retail SME context, focusing on both revenue performance and pricing stability. All three pricing strategies were implemented within a controlled simulation framework using consistent datasets, demand models, and evaluation procedures. This ensured that differences in performance could be attributed directly to pricing logic rather than inconsistencies in implementation.

The findings demonstrate that **no single pricing strategy universally outperformed the others** across all evaluation dimensions. The **machine learning model achieved the highest revenue performance**, confirming its effectiveness in optimisation-driven pricing. However, this was accompanied by the highest levels of pricing volatility. In contrast, the hybrid model achieved the strongest pricing stability outcomes by incorporating constraint-based adjustment mechanisms that moderated price variation over time. Although the hybrid approach produced slightly lower revenue outcomes, the reduction remained comparatively moderate relative to the improvement in pricing consistency.

The rule-based model produced intermediate results across both revenue and stability dimensions. While its deterministic structure provided interpretability and operational simplicity, it did not fully optimise either performance objective. Overall, the findings demonstrate that pricing models reflect their underlying design priorities, with machine learning approaches

favouring optimisation, rule-based systems prioritising deterministic control, and hybrid models balancing adaptability with stability.

In addressing the research question, the study concludes that the machine learning model performs best in terms of revenue generation, whereas the hybrid model performs best in terms of pricing stability. The hybrid strategy also provides the **most balanced overall performance** within retail SME environments. These findings reinforce the importance of evaluating dynamic pricing systems using multiple criteria rather than relying solely on optimisation outcomes.

6.2 Contributions

This study contributes to both academic research and practical pricing strategy evaluation within retail SME contexts. Academically, the research addresses a gap in the literature by comparatively evaluating rule-based, machine learning-based, and hybrid pricing approaches within a consistent experimental framework. Existing studies frequently examine these paradigms separately or within data-rich environments, whereas this study evaluates them under shared SME-focused conditions.

A key contribution is the explicit integration of **pricing stability into the evaluation framework**. Much of the dynamic pricing literature prioritises revenue or profit maximisation as the primary indicator of model effectiveness. By incorporating metrics related to adjustment magnitude, price dispersion, and maximum price variation, the study provides a broader perspective on pricing system performance.

The study also contributes methodologically through the development of a reproducible computational framework for comparative pricing analysis. All pricing strategies were

implemented using identical datasets, demand models, candidate price sets, and evaluation procedures within a deterministic simulation pipeline. The inclusion of statistical testing further strengthens the reliability of the comparative findings.

From a practical perspective, the findings demonstrate that hybrid pricing approaches can provide a balanced relationship between optimisation performance and pricing stability, making them particularly suitable for resource-constrained retail SME environments.

6.3 Recommendations

Several recommendations emerge from the findings of this study. First, retail SMEs should avoid evaluating pricing systems solely according to revenue maximisation outcomes. Although optimisation-driven pricing models may improve financial performance, they can also introduce substantial pricing volatility that may affect operational consistency and customer perception.

The findings suggest that **hybrid pricing models provide the most balanced performance** across the evaluated dimensions. By combining adaptive optimisation with explicit constraints on pricing behaviour, hybrid systems can maintain competitive revenue outcomes while reducing excessive variability. SMEs seeking to adopt dynamic pricing systems should therefore consider hybrid approaches as a practical alternative to fully unconstrained optimisation models.

For organisations with limited technical capability or restricted historical data, rule-based pricing systems may still provide a useful starting point due to their transparency and ease of implementation. However, SMEs relying on such systems may benefit from gradually incorporating data-driven components as analytical capability and data availability improve.

The study also highlights the importance of aligning pricing systems with organisational readiness. Machine learning pricing approaches depend heavily on reliable data quality, monitoring capability, and ongoing model maintenance. SMEs should therefore assess their technical infrastructure carefully before implementing highly adaptive pricing systems.

Finally, organisations should adopt multi-dimensional evaluation frameworks when assessing pricing strategy performance. Stability measures such as adjustment magnitude, price dispersion, and maximum price fluctuation provide additional insight into how pricing behaviour evolves over time and support more balanced decision-making.

6.4 Future Work

This study provides several opportunities for future research. One direction is the use of more advanced demand modelling techniques. The current study relies on a linear regression model, which may not capture complex or non-linear demand relationships. Future research could explore more advanced machine learning or deep learning approaches to improve predictive performance.

Another important extension involves real-world deployment or online experimentation. Although offline simulation provides a controlled environment for comparative analysis, it does not capture direct customer responses to pricing changes. Future studies could evaluate pricing strategies within live or experimental retail settings to examine behavioural effects and long-term outcomes.

Future work could also extend the analysis to additional products, markets, and datasets in order to improve generalisability. Incorporating factors such as competitor pricing, inventory

constraints, and promotional strategies would further strengthen the realism of pricing simulations.

Finally, future research could examine customer perception and fairness more directly. While this study used pricing stability as a proxy for customer-oriented pricing behaviour, behavioural experiments or survey-based approaches could provide deeper insight into how pricing strategies influence customer trust and fairness perceptions.

References

- Bansal, A., Sharma, R. and Kathuria, M. (2022) ‘A Systematic Review on Data Scarcity Problem in Deep Learning: Solution and Applications’, *ACM Computing Surveys*, 54(10). Available at: <https://doi.org/10.1145/3502287>
- Chen, B. *et al.* (2022) ‘Dynamic Pricing and Inventory Control with Fixed Ordering Cost and Incomplete Demand Information’, *Management Science*, 68(8), pp. 5684-5703. Available at: <https://doi.org/10.1287/mnsc.2021.4171>
- Chen, X. and Wang, Y. (2023) ‘Robust Dynamic Pricing with Demand Learning in the Presence of Outlier Customers’, *Operations Research*, 71(4), pp. 1362-1386. Available at: <https://doi.org/10.1287/opre.2022.2280>
- Chen, X., Simchi-Levi, D. and Wang, Y. (2022) ‘Privacy-Preserving Dynamic Personalized Pricing with Demand Learning’, *Management Science*, 68(7), pp. 4878-4898. Available at: <https://doi.org/10.1287/mnsc.2021.4129>
- Cohen, M.C., Elmachoub, A.N. and Lei, X. (2022) ‘Price Discrimination with Fairness Constraints’, *Management Science*, 68(12), pp. 8536-8552. Available at: <https://doi.org/10.1287/mnsc.2022.4317>
- Cohen, M.C., Miao, S. and Wang, Y. (2025) ‘Dynamic Pricing with Fairness Constraints’, *Operations Research*, 73(6), pp. 3027-3043. Available at: <https://doi.org/10.1287/opre.2023.0123>

Das, P. *et al.* (2024) ‘Optimizing Real-Time Dynamic Pricing Strategies in Retail and E-Commerce Using Machine Learning Models’, *The American Journal of Engineering and Technology*, 6(12), pp. 163-177. Available at: <https://doi.org/10.37547/tajet/Volume06Issue12-15>

Deng, G.F. (2023) ‘Dynamic price competition market for retailers in the context of consumer learning behavior and supplier competition: Machine learning-enhanced agent-based modeling and simulation’, *Advances in Production Engineering & Management*, 18(4), pp. 434-446. Available at: <https://doi.org/10.14743/apem2023.4.483>

den Boer, A.V. and Keskin, N.B. (2022) ‘Dynamic Pricing with Demand Learning and Reference Effects’, *Management Science*, 68(10), pp. 7112-7130. Available at: <https://doi.org/10.1287/mnsc.2021.4234>

Diao, W., Harutyunyan, M. and Jiang, B. (2023) ‘Consumer Fairness Concerns and Dynamic Pricing in a Channel’, *Marketing Science*, 42(3), pp. 569-588. Available at: <https://doi.org/10.1287/mksc.2022.1395>

Fan, J., Guo, Y. and Yu, M. (2024) ‘Policy Optimization Using Semiparametric Models for Dynamic Pricing’, *Journal of the American Statistical Association*, 119(545), pp. 552-564. Available at: <https://doi.org/10.1080/01621459.2022.2128359>

Gore, S. (2026) *Dynamic Pricing Study* [GitHub repository]. Available at: https://github.com/shraddha-gore/dynamic_pricing_study (Accessed: 08 May 2026).

Gouveia, F.D. and Mamede, H.S. (2022) ‘Digital Transformation for SMES in the Retail Industry’, *Procedia Computer Science*, 204, pp. 671-681. Available at: <https://doi.org/10.1016/j.procs.2022.08.081>

Immadisetty, A. (2025) 'Dynamic Pricing Strategies in Retail: Leveraging Real-Time Data Analytics for Competitive Advantage', *Journal of Recent Trends in Computer Science and Engineering*, 13(1), pp. 53-65. Available at: <https://doi.org/10.70589/JRTCSE.2025.13.1.8>

Jacobsen, G.D. and Stewart, J.I. (2022) 'How do consumers respond to price complexity? Experimental evidence from the power sector', *Journal of Environmental Economics and Management*, 116. Available at: <https://doi.org/10.1016/j.jeem.2022.102716>

Jullien, B., Reisinger, M. and Rey, P. (2023) 'Personalized Pricing and Distribution Strategies', *Management Science*, 69(3), pp. 1687-1702. Available at: <https://doi.org/10.1287/mnsc.2022.4437>

Kabilan (no date) *Online Retail II Dataset*. Kaggle. Available at: <https://www.kaggle.com/datasets/kabilan45/online-retail-ii-dataset> (Accessed: 08 May 2026).

Kayikci, Y. *et al.* (2022) 'Data-driven optimal dynamic pricing strategy for reducing perishable food waste at retailers', *Journal of Cleaner Production*, 344. Available at: <https://doi.org/10.1016/j.jclepro.2022.131068>

Kigo, S.N., Omondi, E.O. and Omolo, B.O. (2023) 'Assessing predictive performance of supervised machine learning algorithms for a diamond pricing model', *Scientific Reports*, 13(1), pp. Available at: <https://doi.org/10.1038/s41598-023-44326-w>

Kopalle, P. *et al.* (2023) 'Dynamic pricing: Definition, implications for managers, and future research directions', *Journal of Retailing*, 99(4). Available at: <https://doi.org/10.1016/j.jretai.2023.11.003>

Kumari, A., Kumar, S.M. and K, B.R. (2023) 'A Systematic Requirement Analysis for Dynamic Pricing in Retail E-Commerce', *International Journal for Research in Applied Science & Engineering Technology*, 11(3), pp. 342-346. Available at:

<https://doi.org/10.22214/ijraset.2023.49418>

Li, D. and Xin, J. (2024) 'Deep learning-driven intelligent pricing model in retail: from sales forecasting to dynamic price optimization', *Soft Computing*, 28(20), pp. 12281-12297. Available at: <https://doi.org/10.1007/s00500-024-09937-z>

Li, X. and Zheng, Z. (2024) 'Dynamic Pricing with External Information and Inventory Constraint', *Management Science*, 70(9), pp. 5985-6001. Available at:

<https://doi.org/10.1287/mnsc.2023.4963>

Neubert, M. (2022) 'A Systematic Literature Review of Dynamic Pricing Strategies', *International Business Research*, 15(4). Available at: <https://doi.org/10.5539/ibr.v15n4p1>

Nouri-Harzvili, M. and Hosseini-Motlagh, S.M. (2023) 'Dynamic discount pricing in online retail systems: Effects of post-discount dynamic forces', *Expert Systems With Applications*, 232. Available at: <https://doi.org/10.1016/j.eswa.2023.120864>

Nunan, D. and Di Domenico, M.L. (2022) 'Value creation in an algorithmic world: Towards an ethics of dynamic pricing', *Journal of Business Research*, 150, pp. 451-460. Available at: <https://doi.org/10.1016/j.jbusres.2022.06.032>

Peterson, A. (2024) 'Evaluating the Impact of Price Fluctuations on Consumer Buying Decisions and Adaptation Strategies in the Automotive Market', *Law and Economy*, 3(4), pp. 1-9. Available at: <https://doi.org/10.56397/LE.2024.04.01>

- Rahman, M.A. *et al.* (2024) ‘Advancements in Retail Price Optimization: Leveraging Machine Learning Models for Profitability and Competitiveness’, *Journal of Business and Management Studies*, 6(3), pp. 103-110. Available at: <https://doi.org/10.32996/jbms.2024.6.3.11>
- Sarkar, M. *et al.* (2023) ‘Optimizing E-Commerce Profits: A Comprehensive Machine Learning Framework for Dynamic Pricing and Predicting Online Purchases’, *Journal of Computer Science and Technology Studies*, 5(4), pp. 186-193. Available at: <https://doi.org/10.32996/jcsts.2023.5.4.19>
- Shin, D., Vaccari, S. and Zeevi, A. (2023) ‘Dynamic Pricing with Online Reviews’, *Management Science*, 69(2), pp. 824-845. Available at: <https://doi.org/10.1287/mnsc.2022.4387>
- Tan, J. *et al.* (2025) ‘Price fairness perception on online food service platforms: A data-driven approach using fsQCA and machine learning’, *International Journal of Hospitality Management*, 125. Available at: <https://doi.org/10.1016/j.ijhm.2024.104014>
- Wang, X., Liu, S. and Yang, T. (2023) ‘Dynamic pricing and inventory control of online retail of fresh agricultural products with forward purchase behavior’, *Economic Research-Ekonomika Istraživanja*, 36(1). Available at: <https://doi.org/10.1080/1331677X.2023.2180410>
- Yang, N. and Zhang, R. (2022) ‘Dynamic pricing and inventory management in the presence of online reviews’, *Production and Operations Management*, 31(8), pp. 3180-3197. Available at: <https://doi.org/10.1111/poms.13744>
- Zhou, Q. *et al.* (2023) ‘Joint pricing and inventory control with reference price effects and price thresholds: A deep reinforcement learning approach’, *Expert Systems with Applications*, 233. Available at: <https://doi.org/10.1016/j.eswa.2023.120993>

Bibliography

ChemCoder (2025) *How to train AI ML models? Full pipeline in 15 mins*. Available at:

<https://www.youtube.com/watch?v=rysdr4khB5k> (Accessed: 08 May 2026).

GeeksforGeeks (2026) *Types of Machine Learning*. Available at:

<https://www.geeksforgeeks.org/machine-learning/types-of-machine-learning/> (Accessed: 08 May 2026).

pandas (2026) *User Guide*. Available at:

https://pandas.pydata.org/docs/user_guide/index.html#user-guide (Accessed: 08 May 2026).

Pangam, R. (2026) *Guide to the Difference Between Rule-Based and Machine Learning-Based Decision Systems*. Available at:

<https://www.rapidcanvas.ai/guides/guide-to-the-difference-between-rule-based-and-machine-learning-based-decision-systems> (Accessed: 08 May 2026).

scikit-learn (2026) *User Guide*. Available at: https://scikit-learn.org/stable/user_guide.html

(Accessed: 08 May 2026).

Sharma, A. (2026) *Guide for Machine Learning: Types of Machine Learning Algorithms*.

Available at:

<https://medium.com/@akshatsharma0610/guide-for-machine-learning-types-of-machine-learning-algorithms-44a822bfc40c> (Accessed: 08 May 2026).

Streamlit (2026) *Get started with Streamlit*. Available at: <https://docs.streamlit.io/get-started>

(Accessed: 08 May 2026).